Karthik Kannan

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LinkedIn | Portfolio

Ambitious aerospace engineer who enjoys tackling difficult problems, learning unfamiliar systems, and taking ideas from concept to working hardware. Motivated by challenging work that demands creativity, persistence, and technical ownership. Passionate about aerospace and robotics, with a hands-on approach to building, testing, and improving systems.

Professional Experience

Boeing | GNC Engineer 2 | Everett, WA | Apr 2026 - Present

- Verify 777X Primary Flight Function (PFF) requirements through **hardware-in-the-loop (HIL)**, **software-in-the-loop (SIL)**, and **on-aircraft testing** using the Integrated Test Vehicle (ITV) and Psim simulation model.
- Develop **Psim aircraft scenarios** and a custom **Python parser**, reproducing simulated conditions on **ITV live hydraulic hardware**; export test data to **MATLAB for analysis and reporting**.
- Develop **Python tools and structured reporting templates** for global engineering teams, standardizing test documentation and guiding completion of information for Boeing PFF and **FAA certification reviews**.
- Built a **Python onboarding assistant using a local LLM** to help PFF contractors find getting-started guidance in the team wiki.
- Trace upstream and downstream signals in **Simulink** models to identify requirement-verification evidence, investigate discrepancies, and support corrections to test cases and documentation.
- Contribute across PFF safety analysis, requirements, and test activities; collaborate with the **FAA** and engineering stakeholders to support Boeing 777-9 certification.

Mach Industries | Flight Test Engineer | Huntington Beach, CA | Oct 2025 - Mar 2026

- **Planned and led selected flight tests** for a UAS developed from concept to first flight in **three months**, defining flight plans, coordinating execution, and documenting discrepancies and lessons learned.
- Designed and operated propulsion and avionics ground test infrastructure, including **engine test stands**, hot-fire and thermal-soak testing, thrust measurements, and vibration characterization.
- Supported testing of aircraft communications, onboard video streaming, and accelerometer data streams used in **GPS-denied navigation** development; performed post-test data reduction and **root cause analysis**.
- Delivered **technical and product briefings** to senior government officials, military leaders, and international delegations, adapting engineering information to varied audiences. Interviewed prospective engineers.

Collier Aerospace | Structural/Software Engineer | Raleigh, NC | May 2024 - Oct 2025

- Analyzed and sized composite and metallic aerospace structures using **HyperX**, performing trade studies and delivering technical reports for customers including NASA, Electra, ULA, Blue Origin, and Boeing.
- Performed finite element analysis (FEA) in **FEMAP** and **NASTRAN** to evaluate load paths, stress distributions, and potential failure modes in aerospace components.
- Developed **Python automation tools and plug-ins** implementing structural mechanics methods to assess aircraft-specific **spar cutout sizes and shapes**, structural performance, and weight-reduction trade studies.
- Supported environmental loads analysis and **high-temperature composites research**, preparing structural trade studies and technical documentation tailored to customer-specific requirements.

Selected Projects

Poseidon UAV | Senior Design Project Manager | 2023–2024: Led avionics, payload integration, and structural analysis for a NASA Blue Skies disaster-response UAV. Developed **computer vision for landing-site selection and aerial human detection**, evaluated using recorded imagery and video; authored an AIAA paper on disaster-response drone concepts.

GoAERO | Structures and AI/ML Team Lead | 2024–2025: Led a **seven-member team** in structural design and AI/ML integration, including object detection, for a rescue UAV; served as team spokesperson. The team received **~$70,000 during tenure**.

REACHR | Controls, Avionics and Structures Team Lead | 2024–2025: Led a **five-member team** in structures, controls, and avionics for a FEMA-backed amphibious UAV awarded **$80,000**. Adapted **real-time human detection and landing-site selection** to process aircraft streams on a **ground computer** due to onboard compute/storage limits. Supported system architecture and flight controls; prepared SWOT analysis for SBIR proposals.

Materials Research | NC State University | 2022: Developed a **MATLAB/Python neural network** under Dr. Cheryl Xu to predict composite return loss and explore ceramic compositions; transferred the model to an industry collaborator for training on a larger dataset. Received a **$4,000 research grant**.

Personal Robotics Projects: Prototype a **volleyball setting machine** and develop a **wind turbine blade cleaning drone** concept, building skills in mechanical design, electrical systems, software, and integration.

Education

North Carolina State University | Raleigh, NC
Master's in Aerospace Engineering | May 2025 | GPA: **3.9/4.0**
Bachelor's in Aerospace Engineering | May 2024 | GPA: **3.9/4.0**
Minors: Computer Programming (May 2024); Materials Science and Engineering (MSE) (May 2023)

Technical Skills

Software and AI: Python, C/C++, Git, OpenCV, PyTorch, TensorFlow, neural networks, AI/ML, Jenkins, SQL, Java, VBA, JavaScript, HTML/CSS, data parsing, SAS, data recovery, Microsoft Office
Controls and Systems: MATLAB, Simulink, dynamic modeling, UAV autonomy, ArduPilot, Pixhawk, systems integration, HIL/SIL, ITV, Psim, testbed development, DOORS, requirements verification, FMEA, root cause analysis, FAA certification support
Design and Analysis: SolidWorks, Onshape, CATIA, HyperX, FEMAP, NASTRAN, HyperMesh, ANSYS, FEA, fluid dynamics, materials design, technical documentation and report writing